

Time Series Case Study

Data description for a forecasting time series case study with banana prices in the
United Kingdom

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1. Summary

Background: Forecasting time series is a common modelling problem across many domains. Although I have done lots of statistical and machine learning modelling in university, I have mostly focused on cross-sectional methodologies. To address this gap, this report is the first in a series of reports documenting me learning time series forecasting methodologies. This study will aim to revisit an old personal project with a more appropriate time series methodology.

Methods: The project aims to use a linear regression model to forecast weekly banana prices in the United Kingdom using publicly available data collected between 1995 to 2026. Models containing solely features derived from the response of interest (capturing season and trend) will be compared to models containing lagged features derived from the “Origin” variable.

Objectives: The three main objectives of this revisitation are to educate myself on appropriate time series methodologies, build a linear regression model to forecast the average weekly banana price and to post updates as I learn new content. This report aims to provide a description of the project dataset, outline the intended project and provide the first project post on my portfolio.

Deliverable: Interim deliverables will consist of each post related to the project with a final deliverable being a consolidated write-up summarising notable findings.

Value: The value-add of the project is three-fold. Firstly, I will have a real project dataset to work on rather than contrived example datasets. Secondly, I will have personal projects to start showcasing on my portfolio. Finally, the selected model will roll over to a new project where I will aim to deploy the model and assess its performance as new data is published throughout the remainder of the year.

Project timeline: There is no clear timeline, however it is unlikely that this project will not be concluded before the end of 2026. Updates will likely be sporadic given that I am doing this as a personal project.

2. Data description

The following section describes the data employed in this study. The analysis dataset used is the “Machine-readable: Wholesale banana prices weekly 1995 to 2026” comma-separated values (CSV) file. There are additional Open Document Spreadsheet (ODS) files such as “Wholesale banana prices, weekly time series 1995 to 2026” which provide individual worksheets per calendar year with contextual information about the series in each worksheet. The CSV file used for analysis is not a one-to-one representation of these ODS files. The ODS files contain historical context and implies changes in data collection over time.

2.1 Data collection

The data used in this study was obtained from the United Kingdom Governmental website. It was collected by the United Kingdom Department for Environment, Food & Rural Affairs (Defra). Defra describes the series as national averages of the most usual prices charged at wholesale markets in Birmingham and London; the publication is currently updated fortnightly between the years 1995 and 2026 (Department for Environment, Food and Rural Affairs, 2026). The prices are national averages of the most usual prices charged for bananas at wholesale markets in Birmingham and London. Each price is classified under country of origin and origin group. Table 1 outlines the origin group mapping. The origin group is the class label in the data the reference regions are the regions where bananas are originated.

Table 1: Origin group to reference region mapping

Origin group	Reference regions
ACP	African, Caribbean and Pacific (ACP) countries: Belize, Cameroon, Dominican Republic, Ghana, Ivory Coast, Jamaica, Surinam, Windward Islands (ie St Lucia, the Grenadines, Dominica & Grenada), Somalia and St Vincent
Dollar	Colombia, Costa Rica, Ecuador, Honduras, Mexico, Panama, Guatemala, Nicaragua, Malaysia, Venezuela and Brazil
EU	Martinique and Guadeloupe
All	All bananas

The ODS worksheets up to 2018 employ a different wide-format structure containing aggregate origin categories, country-specific series, supply codes, and lower, average and upper most-usual prices. From 2019, the workbook uses a simplified country-by-week structure centred on the average most usual price and a country-group classification.

2.2 Measurement

The project dataset provides average wholesale prices of bananas by country of origin, the country of origin and the day, month, year collected. Prices are in average pounds (£) per kg. The response of interest is the most usual price of bananas across all regions. It seems that when transitioning from 2018 to 2019 a new naming format was adopted in the analysis dataset. Until 2018 the origin label for the response of interest was “all_bananas_bananas” . From January 2019 onwards, the “all_bananas” origin label was used. Additionally, in

December 2024, Defra appears to have changed data collection from weekly to fortnightly, although the workbook continues to represent the time dimension using weekly date columns.

2.3 Statistical techniques

Linear regression will be used to model a linear additive relationship between predictor variables and the average banana price. While the relationship may not be linear in long-term horizons such as year-to-year, this research aims to evaluate whether short-term inter-week forecasting can be appropriately approximated as a linear relationship. This research aims to use the methodologies proposed by Hyndman and Athanasopoulos (2021) to study the time series.

3. References

Department for Environment, Food and Rural Affairs, 2026. *Imported banana prices at Birmingham and London wholesale markets*. [Online]

Available at: <https://www.gov.uk/government/statistical-data-sets/banana-prices>

[Accessed September 2026].

Hyndman, R. J. & Athanasopoulos, G., 2021. *Forecasting: principles and practice*. 3rd ed. Melbourne: OTexts.